

Database Management Systems

CPS 3740

Lecture 08

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Outline

- Continue last lecture
- HTML input format and handling
- Project - SQL Injection
- PHP MySQL error handling
- Midterm 1

HTML inputs

- HTML input allows a user to enter data and use HTML FORM to pass to a CGI program.
- These HTML input types are often listed on the data entry forms.
- Radio buttons: let a user select only one of a limited number of choices
- Checkbox: let the user select one or more options of a limited number of choices.
- Dropdown: let user select one or more options from a drop-down list with options
- Textarea: let user enter a multi-line text input

HTML input types

- These HTML input must be between <FORM> and </FORM>
- <input type="button">
- <input type="checkbox">
- <input type="color">
- <input type="date">
- <input type="datetime-local">
- <input type="email">
- <input type="file">
- <input type="hidden">
- <input type="image">
- <input type="month">
- <input type="number">
- <input type="password">
- <input type="radio">
- <input type="range">
- <input type="reset">
- <input type="search">
- <input type="submit">
- <input type="tel">
- <input type="text">
- <input type="time">
- <input type="url">
- <input type="week">

https://www.w3schools.com/html/html_form_input_types.asp

Retrieve HTML input types in PHP

- These inputs should be all required.
 required>
- <input type="date" name="mydate">
 \$date=\$_POST["mydate"];
- <input type="search">
- <input type="text">
- <input type="hidden">
- <input type="checkbox">
- <input type="file"> → useful, upload, take CPS 3525

http://vader.kean.edu/CPS3740/input_types.html

Check HTML input types in PHP

- Always check if the index key is defined (has value)
if (!isset(\$_POST["mydate"]))
 echo "
mydate is not set\n";
else {
 \$mydate=\$_POST["mydate"];
 echo "
mydate: \$mydate\n";
}

HTML radio buttons and PHP

- Example of HTML Code:

Please select your gender:

<input type="radio" name="gender" value="male">Male

<input type="radio" name="gender" value="female">Female

<input type="radio" name="gender" value="other">Other

Please select your gender:
☐ Male ☐ Female ☐ Other

- Corresponding PHP code to handle the radio button

```
if (empty($_POST["gender"])) {  
    $genderErr = "Gender is required";  
} else {  
    $gender = $_POST["gender"];  
}
```

https://www.w3schools.com/tags/att_input_type_radio.asp

HTML checkbox and PHP

- Example of HTML Code:

What vehicle(s) do you have?

<input type="checkbox" id="vehicle1" name="vehicle[]" value="Bike">Bike

<input type="checkbox" id="vehicle2" name="vehicle[]" value="Car">Car

<input type="checkbox" id="vehicle3" name="vehicle[]" value="Boat">Boat

What vehicle(s) do you have?

☐ Bike ☐ Car ☐ Boat

- Corresponding PHP code to handle the checkbox

```
$vehicle = $_POST['vehicle'];
```

```
if(empty($vehicle))
```

```
    echo("You didn't select any vehicle.");
```

```
else {
```

```
    echo "<br>You selected the following vehicle:\n";
```

```
    for($i=0; $i < count($vehicle); $i++)
```

```
        echo( $vehicle[$i] . " ");
```

```
}
```

https://www.w3schools.com/tags/att_input_type_checkbox.asp

HTML textarea and PHP

Please write your comments: (required)

- Example of HTML Code:

```
Please write your comments: (required)<br>
<textarea rows="4" cols="40" name="comment" required>
</textarea>
```

- Corresponding PHP code to receive the comment

```
if (empty($_POST["comment"])) {
    $commentErr = "Comment is required";
} else {
    $comment= $_POST["comment"];
}
```

https://www.w3schools.com/tags/att_textarea_required.asp

HTML dropdown list and PHP - single

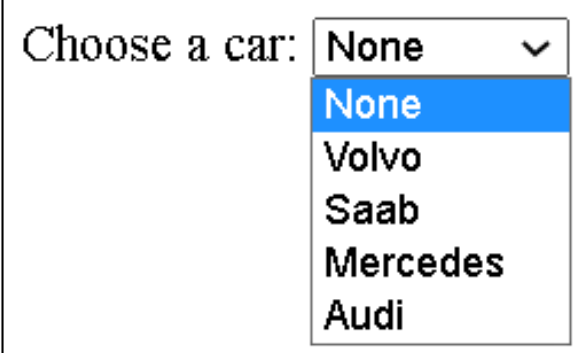
- Example of HTML Code: (single selection)

```
<label for="cars">Choose a car:</label>
<select name="cars" id="cars" required>
  <option value="">None</option>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="mercedes">Mercedes</option>
  <option value="audi">Audi</option>
</select>
```

- Corresponding PHP code to receive the selection

```
if (empty($_POST["cars"])) {
  $carErr = "Car selection is required";
} else {
  $car= $_POST["cars"];
}
```

https://www.w3schools.com/tags/tag_select.asp



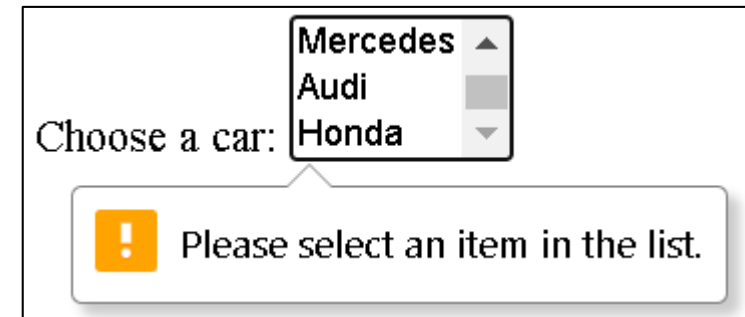
HTML dropdown list and PHP - multiple

- Example of HTML Code: (multiple selections)

```
<label for="cars">Choose a car:</label>
<select name="cars[]" size=3 multiple required>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="mercedes">Mercedes</option>
  <option value="audi">Audi</option>
  <option value="honda">Honda</option>
</select>
```

- Corresponding PHP code to receive the selection

```
if(isset($_POST["cars"])) {
    foreach ($_POST['cars'] as $cars)
        print "<br>You selected $cars \n";
} else
    echo "<br>Select an option first.";
```



https://www.w3schools.com/tags/att_select_multiple.asp

Example: display products

- Retrieve data from the specific table/database
- Inside the while loop
 - Identify the Quantity < 5, make the number in red
 - Calculate profit for every item, and sum the profit
- Outside the loop, print the Total profit

The following products were match search keyword * .

ID	Name	Description	Type	Cost	Sell price	Quantity	Vendor Name	Added by employee
1	iPod	hi	furniture	2	490	2000	XYZ3	Victor Smith
34	new1	this is a test 2e	electronics	40	250	400	XYZ3	John Lee
39	hi	a new test	electronics	12	14	12	XYZ3	John Lee
40	Apple	Laptoppp	kitchen	15	25	200	XYZ3	John Lee
66	new	this is a new test	furniture	100	333	3	QQQ	Victor Smith

Total profit: USD 1,062,723.00

HTML Exercise – test.html

- Make a web page
 - Under your **public_html/CPS3740** folder
 - A statement displays your name in red color
 - Welcome to **YOUR_NAME**
 - A dropdown lists the course you are taking now
 - An input box for people to enter any text
 - A table show your class schedule with a row header Monday to Friday.
 - 4 checkboxes list the popular sports.
 - 2 radio buttons for people to select major CS or IT
- Every input tag should have unique attribute name
- Change your HTML file permission mode to 705
- Check your web page: (xxxx is your web server user id)
`http://{webserver}/~xxxx/CPS3740/test.html`

SQL Injection

- A SQL injection is a security concern which could be an attack that consists of an **insertion** or "**injection**" of a SQL query through the input data from the client to the application.
- An input box is possible to have SQL injection

Login ID:

Password:

- What happen if a user types
 - Login ID:
 - Password: a' OR 'a'='a

HTML code – logintest.html

```
<HTML>
```

```
<form name="input" action=" logintest.php" method="post" >
```

```
<br> Login ID: <input type="text" name="login">
```

```
<br> Password: <input type="password" name="password">
```

```
<input type="submit" value="Login">
```

```
</form>
```

```
</HTML>
```

<http://vader.kean.edu/CPS3740/logintest.html>

PHP code – logintest.php

```
<?php
echo "<HTML>\n";
include("dbconfig.php");
$con = mysqli_connect($hostname,$username,$password,$dbname);
$login=$_POST["login"];
$password=$_POST["password"];
$sql= "SELECT login FROM dreamhome.Users WHERE login='$login' AND
password='$password'";
echo "<br>$sql\n"; // debug purpose, remove it after complete test code
$result = mysqli_query($con, $sql);
if ($result ) {
    $num=mysqli_num_rows($result);
    if ($num>0) {
        echo "<br>Login successfully!\n";
    }
    else {
        echo "<br>Login failed!\n";
    }
}
?>
```


SQL Injection Risks 1

Login ID:

Password: a' OR 'a'='a

```
$login=$_POST["login"];  
$bpassword=$_POST["password"];  
$sql= "SELECT login FROM Users WHERE  
      login='$login' AND password='$bpassword';
```

```
$sql= "SELECT login FROM Users WHERE  
      login='' AND password= 'a' OR 'a'='a' ";  
// false AND false OR true  
//      false      OR true  
// final result: true => what happen?  
// all the records will be selected. -> $num>0 -> login successful
```

How to Fix SQL Injection?

- PHP provides function `mysqli_real_escape_string()` to escape special characters in a string.

```
$login=mysqli_real_escape_string($con,$_POST["login"]);  
$bpassword=mysqli_real_escape_string($con,$_POST["password"]);  
$query = "SELECT login FROM Users WHERE  
        login='$login' AND password='$password'";
```

- Login_ID:
- Password: a' OR 'a'='a

```
$query = "SELECT login FROM Users WHERE  
        login= '' AND password='a\' OR \'a\'=\'a'";
```

Hacking Exercise – preparation

- Create a table Users_XXXX in class database (eg: CPS3740_2021F, TECH3740_2021S1)

```
mysql> CREATE table Users_XXXX (login varchar(20), password  
varchar(20),name varchar(20));
```

--- XXXX is your database login

- Get permission on Users_XXXX on the class website, opt 6.3
- Insert a record:

```
mysql> INSERT into Users_XXXX  
values ('tester01', 'test01','John');
```

Hacking Exercise – SQL injection Risk1

Risk1: Unauthorized users can login successfully.

1. Copy logintest.html(from slide) to injection1.html
2. Copy logintest.php(from slide) to injection1.php
3. Change injection1.html action="injection1.php"
4. injection1.php: modify the following SQL statement
 - Change table name from dreamhome.Users to <classDB>.Users_xxxx
(eg: TECH3740_2022S.Users_demo01.....)
5. Follow the previous slide – SQL Injection Risks 1
 - Test case#1: login with correct password('tester01', 'test01')
 - Test case#2: login with wrong password ('tester01', '111')
 - Test case#3: login with SQL injection 1
Login ID:
Password: a' OR 'a'='a

PS: If your config.php is in different directory, please include the directory path to import the correct config.php file.

Hacking exercise – SQL injection Risk2

Risk2: Unauthorized users can delete your records (drop tables) without login successfully.

- (1) Copy your injection1.html to injection2.html
- (2) Copy your injection1.php to injection2.php
- (3) **Change injection2.html action="injection2.php"**
- (4) injection2.php : replace mysqli_query session with the following mysqli_multi_query session

```
if ( mysqli_multi_query($con,$sql) ) {  
    do { // Store result set  
        if ( $result=mysqli_store_result($con) ){  
            // Store first result set  
            $num=mysqli_num_rows($result);  
            if ($num>0) {  
                echo "<br>Login successfully!\n";  
            }  
            else {  
                echo "<br>Login failed!\n";  
            }  
        }  
    } while ( mysqli_next_result($con));  
}
```

Hacking exercise – SQL injection Risk2(cont.)

(5) Follow the previous slide – SQL Injection Risks 2

- Test case#1:

Login ID:

Password: '; DELETE from **Users_xxxxx** ; --

(1.1)password: **replace xxxx with your dblogin name**
spaces after –

// eg: '; DELETE from Users_demo01 ; --

(1.2) check data:

select * from Users_xxxxx; **// all records are deleted.**

- Test case#2:

Login ID:

Password: '; DROP TABLE **Users_xxxx** ; --

(1.1)password: **replace xxxx with your dblogin name**
spaces after –

// eg: '; DROP TABLE Users_demo01 ; --

(2.2) check data:

select * from Users_xxxxx; **// table does not exist**

(Dangerous) Unauthorized users can delete your records (drop tables) without login successfully.

SQL Injection Risks 2: why?

(1) Test case #1:

(1.1)password:replace xxxx with your dblogin name) (1.2) spaces after --

Login ID:

Password: ' ; DELETE from **Users_xxxxx** ; --

Reason: SELECT login FROM Users WHERE login= "

AND password= ' '; DELETE from <classDB>.Users_xxxx ; -- '

=> All data in Users_xxxx are deleted.

(2) Test case #2:

Login ID:

Password: ' ; DROP TABLE **Users_xxxx** ; --

Reason: `SELECT login FROM Users WHERE login= "`

AND password= ' '; DROP TABLE Users_xxxx ; -- '

=> Users_xxxx is dropped.

(3) How to Fix SQL Injection?

mysqli_real_escape_string()

Protect dbconfig.php

```
<?php
define("IN_CODE", 1);
include("dbconfig.php");
$con = mysqli_connect($dbhostname,$dbusername,$dbpassword,$dbname);
```

```
echo "<HTML>\n";
$login=$_POST["login"];
$password=$_POST["password"];
```

```
$sql= "SELECT login FROM Users WHERE login='$login' AND
password='$password'";
$result = mysqli_query($con, $sql);
$num=mysqli_num_rows($result);
```

```
if ( $num < 1)
{ echo "<br>Login failed."; }
else { echo "<br>Login sucessfull!"; }
mysqli_free_result($result);
mysqli_close($con);
?>
```

```
// dbconfig.php
<?php
if(!defined("IN_CODE")) {
die("not an entry point");
}
...
?>
```


Authentication in PHP (Bad design)

- DO NOT use the number of returned rows for authentication
→ to avoid SQL injection

```
$sql= "SELECT login, password FROM Users WHERE  
login='$login' AND password='$bpassword'";
```

```
$result = mysqli_query($con, $sql);  
$num=mysqli_num_rows($result);
```

```
if ( $num < 1)  
{ echo "<br>Login failed."; }  
else  
{ echo "Login successful!"; }
```

Class exercise - login

- Based on display program. Must use POST method

```
<?php
```

```
error_reporting(E_ALL|E_STRICT); //turn on error display
```

```
$username = $_POST["username"];
```

```
if ($username == "" ) {
```

```
    die(" <br> Please enter username. \n");
```

```
}
```

```
$query="select password, name from TABLE where login='$username';
```

```
#### send the query to the DB and get the result.
```

```
#### loop and if statement for authentication
```

```
{
```

```
    echo "<br>User name and password are not matched.\n";
```

```
    die; ## DO NOT continue! No database connection!
```

```
}
```

```
echo "Welcome user: <b>Test</b>.\n";
```

```
?>
```

Authentication in PHP (better design)

- Avoid SQL injection
- Always check login and password against in database

```
$sql= "SELECT login, password FROM Users WHERE login='$login' " ;  
$result = mysqli_query($con, $sql);  
$num=mysqli_num_rows($result);  
If ($num==0) echo "No such user $login";  
else if ($num>1) echo "More than one user login $login" ;  
else {  
    $row = mysql_fetch_array($result);  
    if ($row['password']==$bpassword)  
    { echo "<br>Login successful."; }  
    else { echo "Login failed (user exists, but wrong password!"; }  
}
```

Class exercise- login with DB, cookie

- 3 passwords variables in the PHP
 - Connection variable to MySQL
 - `$_POST[]` that user enters on the input box
 - The correct password from the user table for the specific login
- Only use the login condition in the SQL statement
- Problem with using both login and password
 - Cannot tell whether user login is in DB or not
- Where is the correct place to set the cookie?
 - After compare the two passwords match
- Students complete the PHP MySQL codes in 5-10 minutes

PHP MySQL – debug errors

- Fix this SQL statement to test your program

```
$sql= "SELECT login password FROM users WHERE  
login='$login ' AND password='$bpassword";
```

```
$result = mysqli_query($con, $sql);
```

- What happens to \$result if SQL has a syntax error?
 - Missing quotes (single or double)
 - Typo or spelling error (table or column)
 - Incorrect syntax (missing “,”, operators, keywords, etc)
- \$result – the difference between SQL error or 0 return.
- Add mysqli_error() in the project
 - Where you have mysqli_query()

Project demo

- Demo page: <http://vader.kean.edu/CPS3740/>
- Features
 - Search → discussed before
 - Insert into table → will discuss later
 - Update records → will discuss later

Project Development Flow

Build basic personal web pages with FORM



Display data from database with Simple PHP

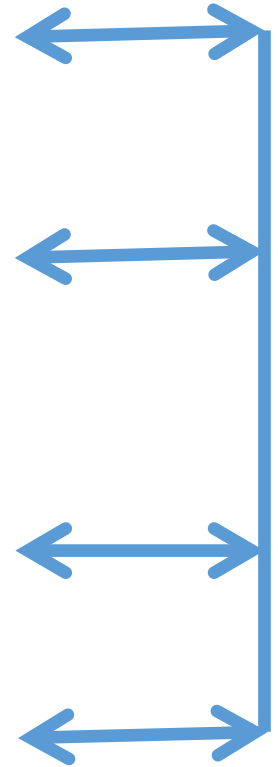


Login, add/search data by FORM, PHP



Advanced queries & PHP

Put the project link on your resume



Refine project

Example 6.26 Three Table Join

- For each branch, list staff who manage properties, including city in which branch is located and properties they manage.

```
SELECT b.branchNo, b.city, s.staffNo, fName, lName,  
       propertyNo
```

```
FROM Branch b, Staff s, PropertyForRent p
```

```
WHERE b.branchNo = s.branchNo AND  
       s.staffNo = p.staffNo
```

```
ORDER BY b.branchNo, s.staffNo, propertyNo;
```

- **Note: join 3 tables, must have at least 2 joining conditions to connect 3 tables.**

Self-study

Example 6.26 Three Table Join

branchNo	city	staffNo	fName	lName	propertyNo
B003	Glasgow	SG14	David	Ford	PG16
B003	Glasgow	SG37	Ann	Beech	PG21
B003	Glasgow	SG37	Ann	Beech	PG36
B005	London	SL41	Julie	Lee	PL94
B007	Aberdeen	SA9	Mary	Howe	PA14

- Alternative formulation for FROM and WHERE:

**FROM (Branch b JOIN Staff s USING branchNo) AS
bs JOIN PropertyForRent p USING staffNo**

Self-study

Class exercise – JOIN 3 tables (ex: Claudia's transcript)

- Use **Students**, **Courses**, **Students_Courses** tables in dreamhome
- List the transcript for Claudia. The output should be the course name, semester, year, credit and grade.
 1. Check the fields for each table
 2. Find the common column names between tables
 3. Join the tables with alias
 4. Make sure all 3 tables are connected.
 5. Narrow down the result with a condition "Claudia"
 6. Select the columns with table alias
 7. Rename the output to have meaningful header

Class exercise – JOIN 3 tables (Answer) (ex: Claudia's transcript)

- Use **Students**, **Courses**, **Students_Courses** tables in dreamhome
- List the transcript for Claudia. The output should be the course name, semester, year, credit and grade.

```
select sc.year, sc.semester, c.name as course_name,  
       c.credits, sc.grade  
from Students_Courses sc, Students s, Courses c  
where sc.sid=s.sid and sc.cid=c.cid  
      and s.first_name='Claudia'  
order by sc.year, sc.semester;
```

year	semester	course_name	credits	grade
2014	Fall	Java2	4	F
2015	Spring	Java2	4	A
2016	Fall	Database Systems	3	C
2016	Spring	Data Mining	3	B

Self-study

Class exercise – JOIN 3 tables (ex. all students' transcripts)

- Use **Students**, **Courses**, **Students_Courses** tables in dreamhome
- List the course count list for every student. The output should be the student first_name, last_name, course name, counts.
- The output should be order by student last name, the year and semester when the course was taken.
 - Use the previous example
 - Remove the condition that has “Claudia”
 - Add group by - But which fields to group by?
 - Add order by
- Verify your output
 - It should have all the records in the Students_Courses table.
 - The output is sorted correctly.

Self-study

Class exercise – JOIN 3 tables (Answer) (ex. all students' transcripts)

<Answer>

```
select s.sid, s.first_name, s.last_name,  
       sc.year, sc.semester, c.name as course_name,  
       c.credits, sc.grade  
from Students_Courses sc, Students s, Courses c  
where sc.sid=s.sid and sc.cid=c.cid  
order by s.sid, s.first_name, s.last_name, sc.year, sc.semester;
```

sid	first_name	last_name	year	semester	course_name	credits	grade
1001	Austin	Huang	2013	Fall	Java2	4	A-
1001	Austin	Huang	2016	Fall	Data Structure	4	C+
1001	Austin	Huang	2017	Spring	Web Programming	3	C+
1007	Sarah	Lee	2014	Fall	Database Introductio	3	A
1003	Claudia	Lee	2014	Fall	Java2	4	F
1003	Claudia	Lee	2015	Spring	Java2	4	A
1007	Sarah	Lee	2015	Spring	Java2	4	B
1003	Claudia	Lee	2016	Fall	Database Systems	3	C
1003	Claudia	Lee	2016	Spring	Data Mining	3	B
1004	Andrew	Lin	2013	Fall	Java2	4	C
1004	Andrew	Lin	2014	Fall	Data Structure	4	C
1004	Andrew	Lin	2014	Spring	Data Structure	4	B+
1004	Andrew	Lin	2014	Spring	Java2	4	A-
1005	Helen	Wu	2016	Spring	Java2	4	A

Self-study

Class exercise – JOIN 3 tables with group by

- List the course count list for every student. The output should be the student first_name, last_name, course name, counts.
- The output should be order by student last name.

```
select s.first_name, s.last_name, c.name as  
course_name, count(s.sid) as counts
```

```
from Students_Courses sc, Students s, Courses c
```

```
where sc.sid=s.sid
```

```
and sc.cid=c.cid
```

```
group by s.sid, c.cid
```

```
order by s.last_name;
```

first_name	last_name	course_name	counts
Austin	Huang	Data Structure	1
Austin	Huang	Web Programming	1
Austin	Huang	Java2	1
Claudia	Lee	Database Systems	1
Sarah	Lee	Java2	1
Claudia	Lee	Data Mining	1
Sarah	Lee	Database Introductio	1
Claudia	Lee	Java2	2
Andrew	Lin	Java2	2
Andrew	Lin	Data Structure	2
Helen	Wu	Java2	1

Self-study

Home exercise - Dropdown list

- For the add transaction page, how to make the source dropdown list
- Use `<SELECT>` to pass the HTML name
- The list items should be in `<OPTION>`
 - Generate list items by using loop and `mysqli_fetch_array()` must be inside the loop
 - Use `<OPTION value='$id'> $name</OPTION>`

Home exercise - Dropdown list

- Using <SELECT>

Select a vendor:

```
<SELECT name='vendor_id'>
```

```
<option value=1001>XXX</option>
```

```
<option value=1002>ABC</option>
```

```
...
```

```
</SELECT>
```

- Highlight in red should be generated by a loop inside:

```
while($row = mysqli_fetch_array($result)){  
    $id = $row['....'];  
    echo "<option value='.... '></option> ";  
}
```


Home Exercise – JOIN 4 tables with aggregation (ex. Claudia's tuition)

- Use **Students**, **Courses**, **Students_Courses**, **Tuitions** tables in dreamhome
- List the total amount Claudia need to pay. The output should be only the amount.
 1. SQL query is similar to the previous one, but need to join 4 tables.
 2. Need to add one more joining condition → what?
 3. **Each course cost = credits * fee_per_credit** for that year
 4. Have the aggregation function (which one?) in the SELECT
 5. Rename the output to have meaningful header

Home Exercise – JOIN 4 tables with aggregation (answer) (ex. Claudia's tuition)

(1) List the cost for each course for Claudia

```
select sc.year, sc.semester, c.name as course_name, c.credits, sc.grade,  
       t.fee_per_credit, c.credits*t.fee_per_credit as cost  
from Students_Courses sc, Students s, Courses c , Tuitions t  
where sc.sid=s.sid and sc.cid=c.cid and t.year=sc.year  
and s.first_name='Claudia'  
order by sc.year, sc.semester;
```

year	semester	course_name	credits	grade	fee_per_credit	cost
2014	Fall	Java2	4	F	140.5	562
2015	Spring	Java2	4	A	150.5	602
2016	Fall	Database Systems	3	C	160.5	481.5
2016	Spring	Data Mining	3	B	160.5	481.5

(2) Just show the total amount Claudia need to pay..

```
select sum(c.credits*t.fee_per_credit) as total  
from Students_Courses sc, Students s, Courses c , Tuitions t  
where sc.sid=s.sid and sc.cid=c.cid and t.year=sc.year  
and s.first_name='Claudia';
```

total
2127

Self study

Home exercise – JOIN 4 tables with aggregation and group by (ex. all students' tuition)

- Use **Students**, **Courses**, **Students_Courses**, **Tuitions** tables in dreamhome
- List the total money each student to pay
- The output should be student first_name, last_name, amount, and sorted by last_name from A to Z.
 - Based on the SQL query in the previous examples
 - Need to join 4 tables
 - Remove the condition that has “Claudia”
 - Need to use group by
 - Have the aggregation function (which one?) in the SELECT
 - Rename the output to have meaningful headers

Home exercise – JOIN 4 tables with aggregation and group by (Answer) (ex. all students' tuition)

(1) List the cost of each course for each student

```
select s.sid, s.first_name, s.last_name,  
       sc.year, sc.semester, c.name as course_name, c.credits, sc.grade,  
       t.fee_per_credit, c.credits*t.fee_per_credit as cost  
from Students_Courses sc, Students s, Courses c , Tuitions t  
where sc.sid=s.sid and sc.cid=c.cid and t.year=sc.year  
order by s.sid, s.first_name, s.last_name, sc.year, sc.semester;
```

Home exercise – JOIN 4 tables with aggregation and group by (Answer) (ex. all students' tuition)

sid	first_name	last_name	year	semester	course_name	credits	grade	fee_per_credit	cost
1001	Austin	Huang	2013	Fall	Java2	4	A-	130.5	522
1001	Austin	Huang	2016	Fall	Data Structure	4	C+	160.5	642
1001	Austin	Huang	2017	Spring	Web Programming	3	C+	200.5	601.5
1003	Claudia	Lee	2014	Fall	Java2	4	F	140.5	562
1007	Sarah	Lee	2014	Fall	Database Introductio	3	A	140.5	421.5
1007	Sarah	Lee	2015	Spring	Java2	4	B	150.5	602
1003	Claudia	Lee	2015	Spring	Java2	4	A	150.5	602
1003	Claudia	Lee	2016	Fall	Database Systems	3	C	160.5	481.5
1003	Claudia	Lee	2016	Spring	Data Mining	3	B	160.5	481.5
1004	Andrew	Lin	2013	Fall	Java2	4	C	130.5	522
1004	Andrew	Lin	2014	Fall	Data Structure	4	C	140.5	562
1004	Andrew	Lin	2014	Spring	Java2	4	A-	140.5	562
1004	Andrew	Lin	2014	Spring	Data Structure	4	B+	140.5	562
1005	Helen	Wu	2016	Spring	Java2	4	A	160.5	642

Home exercise – JOIN 4 tables with aggregation and group by (Answer) (ex. all students' tuition)

(2) Use aggregation function to print the tuitions for each student

```
select s.sid, s.first_name, s.last_name,  
       sum(c.credits*t.fee_per_credit) as total
```

```
from Students_Courses sc, Students s, Courses c , Tuitions t  
where sc.sid=s.sid and sc.cid=c.cid and t.year=sc.year
```

```
group by s.sid
```

```
order by s.last_name, s.first_name;
```

sid	first_name	last_name	total
1001	Austin	Huang	1765.5
1003	Claudia	Lee	2127
1007	Sarah	Lee	1023.5
1004	Andrew	Lin	2208
1005	Helen	Wu	642

SQL error handling & PHP

- `mysqli_error()`
 - https://www.w3schools.com/php/func_mysqli_error.asp
- `mysqli_connect_errno()`
 - https://www.w3schools.com/php/func_mysqli_connect_errno.asp
- `mysqli_report(MYSQLI_REPORT_ERROR)`
 - <http://www.nusphere.com/kb/phpmanual/function.mysqli-report.htm>

PHP exception handling

- try {} catch() {}
 - <https://www.guru99.com/error-handling-and-exceptions.html>
- With mysqli
 - <http://php.net/manual/en/class.mysqli-sql-exception.php>
- More
 - <https://stackify.com/php-try-catch-php-exception-tutorial/>

Conclusion

- Work on project
- Test your project following the requirements